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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Mohammed Razia Alamgir Banu
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Roll Number	2520090139
Branch	CSE-IT
Course Outcome	CO-5

Case Study Topic: Cricket Scoring Service using Counting Sort

Work Done Summary:

In this case study, we developed a Java-based sorting system for cricket delivery records. The system sorts ball-by-ball delivery records using a composite key consisting of **over number** and **ball number**. Since the range of overs and balls is small, the Counting Sort algorithm was chosen for efficient sorting.

Using Java, we implemented stable counting sort to first sort deliveries by ball number and then by over number. This follows the LSD radix-style approach, ensuring that the chronological order of deliveries is maintained.

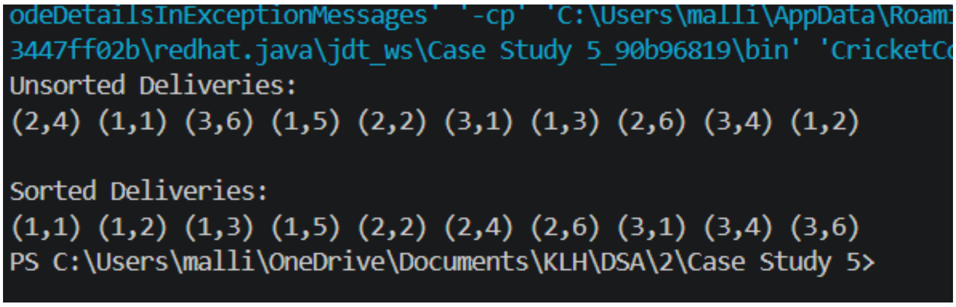
The Counting Sort approach improves performance for large seasonal cricket datasets and provides efficient replay generation and analytical queries.

Implementation Details:

- Created a **Delivery class** in Java to store over number and ball number.
- Stored all delivery records in an array.
- Applied stable Counting Sort on the **ball number** first.
- Applied stable Counting Sort on the **over number** next.
- Used reverse traversal while placing elements to preserve stability.

- Displayed sorted delivery records in chronological order.

Result: Output Screenshots:



```
odeDetailsInExceptionMessages' '-cp' 'C:\Users\malli\AppData\Roami
3447ff02b\redhat.java\jdt_ws\Case Study 5_90b96819\bin' 'CricketC
Unsorted Deliveries:
(2,4) (1,1) (3,6) (1,5) (2,2) (3,1) (1,3) (2,6) (3,4) (1,2)

Sorted Deliveries:
(1,1) (1,2) (1,3) (1,5) (2,2) (2,4) (2,6) (3,1) (3,4) (3,6)
PS C:\Users\malli\OneDrive\Documents\KLH\DSA\2\Case Study 5>
```

Time Complexity:

- Counting Sort: **O(n + k)**
- Two passes: 1. First by ball, 2. Second by over
- Total complexity = O(n+k)

SLA Analysis:

- Total Deliveries = **10**
- Number of Overs = 3
- Maximum Ball Number = 6
- Sorting Passes = 2

GitHub Link: https://github.com/shashankreddy-21/DSA-Project

Student Signature	Faculty Signature